

EQUILIBRIUM SOLID STATE SOLUBILITY AND THE SOLVUS EQUATION OF BINARY SYSTEMS

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Abstract—On the basis of an analysis of experimental data necessary and sufficient conditions have been formulated for obtaining equilibrium experimental curves of solid-state solubility (solvus curves). It is shown that the Vant–Hoff isobar applied to the equilibrium of solid phases in the solvus curve is a linear function for experimental equilibrium curves over a wide range of temperatures (hundreds of degrees). Based on these data the enthalpies of solid-state dissolution in different binary systems have been determined. It is shown that there is an approximately linear dependence of the enthalpy values on the dimensional factor.

The solvus equation of a binary system was obtained and it is shown that the equation is well obeyed by different solid solutions.

Keywords: Solid-state solubility, equilibrium, the solvus equation.

INTRODUCTION

Mutual solubility of components in the subsolidus region has been found in most of the binary systems studied [1, 2]. Almost all of the available solid solubility data were obtained by the sample quenching technique. This method has several shortcomings [3].

(1) The main shortcoming is the impossibility of monitoring the onset of the equilibrium state during the annealing and consequently the great influence of kinetic factors, especially at lower temperatures.

(2) The equilibrium established during the annealing may become shifted upon quenching or in the subsequent study of the composition of the phases. The amount of such a shift depends on the kinetic properties of the particular system.

(3) Quenching induces considerable mechanical strains in the sample affecting their properties. Since quenching usually occurs from different temperatures this results in additional errors in the determination of the composition of the phases.

As a result, the mutual solubility curves obtained by different authors for the components of the same solid solution are in most cases substantially different. The reference literature [1, 2] usually discusses only the degree of agreement between the data of different authors rather than their correspondence to the true equilibrium solvus curve. It is obvious that under normal conditions there will be only one equilibrium solvus curve for each solid solution and even a satisfactory agreement between the experimental data of different authors does not guarantee that they are identical to this equilibrium curve. Up to the present time an estimation of the degree of such correspondence was impossible for the following reasons: (1)

there were no reliable experimental data on the equilibrium solid solubility; and (2) there were no criteria for estimating the degree of equilibrium of the experimental solvus curves.

The problem of equilibration is therefore very important, being in essence the problem of the reliability of a vast amount of experimental data whose obtainment requires a considerable amount of effort. For kinetic reasons the quenching methods are not suitable for the study of equilibrium solid solubility. Considering the above, the necessary and sufficient conditions for obtaining an equilibrium experimental solvus curve can be considered to be:

(1) the possibility of controlling the establishment of equilibrium during annealing; and

(2) determination of the composition of the equilibrium phases directly at the equilibrium temperature.

EXPERIMENTAL

At the present time conditions (1) and (2) are most fully met when using high-temperature X-ray diffraction methods. Earlier we have developed an experimental procedure of this method [4–8] and obtained equilibrium solvus curves for various salt and metallic binary systems. Conditions (1) and (2) were fulfilled as follows:

(1) The control of the equilibrium during the annealing was accomplished in two ways: (a) by means of kinetic curves of time versus crystal lattice parameter and (b) attaining the equilibrium from two sides—from large as well as small concentrations of the components.

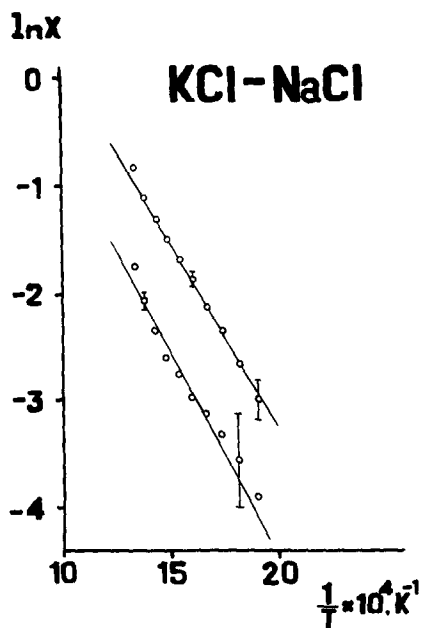


Fig. 1. Vant-Hoff isobar for the equilibrium solvus in the KCl-NaCl system.

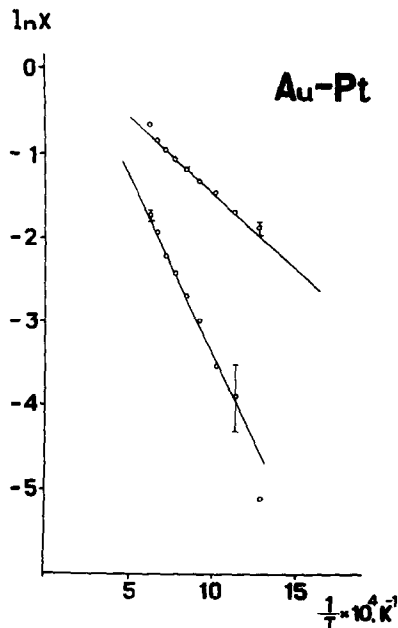


Fig. 3. Vant-Hoff isobar for the equilibrium solvus in the Au-Pt system.

(2) Determination of the composition of the equilibrium phases at the equilibrium temperature was ensured by sufficiently accurate measurements of the lattice parameters of the phases at this temperature using high-temperature Vegard isotherms.

In this way we have obtained equilibrium solvus curves for the KCl-NaCl, CsI-CsBr, Ag-Cu, Au-Pt, Au-Ni, Pb-Sn and Cu-Rh solid solutions. For all of these systems there were solvus curves obtained by the quenching method. A comparison of the

“quenched” and equilibrium curves indicates that the “equilibrium” solubility is usually lower than the “quenched” one. The extent of divergence between these two types of data may be different depending on the kinetic features in particular systems.

RESULTS

The most important feature of the equilibrium solvus curves is the linear form of the dependence

$$\ln X = -\frac{\Delta H_s}{RT} + C, \quad (1)$$

where X is the mole fraction of the dissolved component, ΔH_s is the change in the partial mole enthalpy of the dissolved component as it goes from one phase into another (i.e. the mole enthalpy of dissolution—by analogy with a liquid-solid equilibrium [9]) and C is a constant. Equation (1) is a Vant-Hoff isobar [9] (an integral form) applied to the heterogeneous equilibrium of the decomposition phases.

Figures 1-3 show isobars (1) for the equilibrium solvus curves of the KCl-NaCl, Ag-Cu and Au-Pt solid solutions (according to the data of Refs 4, 6 and 7). In all of these figures the straight line at the top is the solubility of the second component in the first (the left-hand branch of the solvus, see, for example, Fig. 4) and the straight line at the bottom is the solubility of the first component in the second (the right-hand branch of the solvus). Designated by X_1 and X_2 in the solvus curves are the mole fractions of the more soluble component. Therefore in Figs 1-8 $X = X_1$ for the curves at the top and $X = 1 - X_2$ for

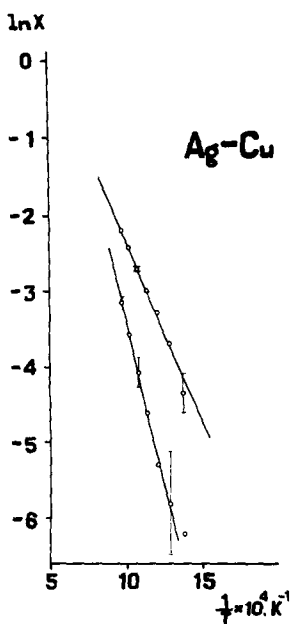


Fig. 2. Vant-Hoff isobar for the equilibrium solvus in the Cu-Ag system.

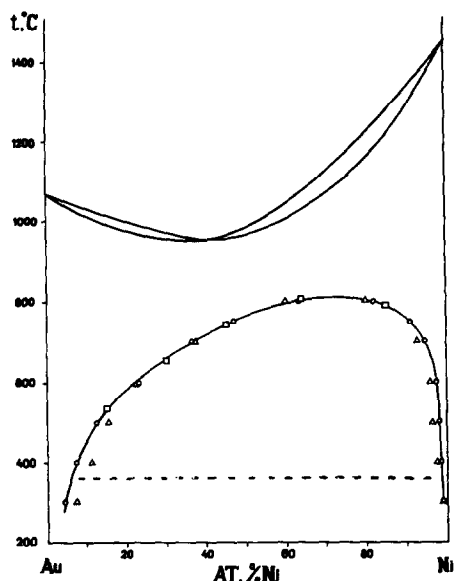


Fig. 4. The equilibrium decomposition curve of the Au-Ni solid solutions according to [8]. Liquidus and solidus as well as Δ -data [1]. The dashed line is the Curie point of nickel.

the curves at the bottom. The vertical line is the maximum error in the determination of the composition ΔX which is a constant within each system and varies from 0.3 at. % for the Ag-Cu system to about 2 mol. % in CsI-CsBr. It can be seen from Figs 1-3 that isobars (1) for the equilibrium mutual solubility curves of these systems are linear within experimental error.

According to (1) the slope of the straight lines in Figs 1-3 determines the enthalpy of dissolution of the components ΔH_s . Table 1 shows the values of ΔH_s for different systems determined from these slopes.

It is obvious that when there is a phase transition in the subsolidus region the values of ΔH_s will be different for different phases and isobar (1) must show deviations from the linearity. As an example Fig. 4 shows an equilibrium solvus curve for the Au-Ni system obtained using the data from Ref. 8. There is a distinct bending in the left-hand branch of the solvus in the vicinity of the Curie point ($\sim 360^\circ\text{C}$) of nickel. Corresponding to this bending is a discontinuity in isobars (1) and a change in their slopes† (see Fig. 5).

As is known, the time required for a solid solution to reach equilibrium is strongly temperature-dependent and is different for different systems. Figure 6 shows the isobars for the solvus branches of the CsI-CsBr system. According to the experimental data [5] this system is characterized by a very small rate at which it achieves equilibrium at temperatures below $+140^\circ\text{C}$. In addition, this rate may be different for each of the two branches. While for the CsI-based

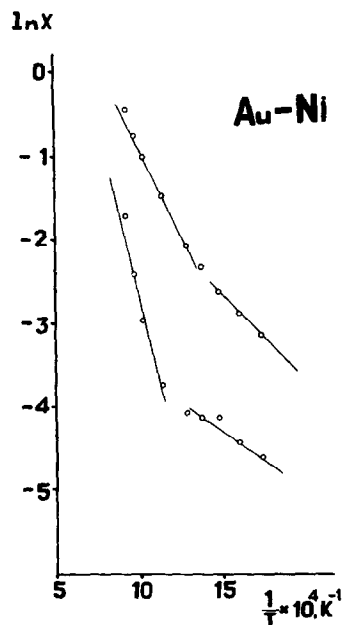


Fig. 5. Vant-Hoff isobars for the branches of the equilibrium solvus in the Au-Ni system.

phase the equilibrium can be considered to have been reached for all of the temperatures studied, the phase based on CsBr does not appear to be in equilibrium at temperatures below $+140^\circ\text{C}$. Accordingly, the isobar of the CsI-based phase (the top line in Fig. 6) is a straight line whereas that of the CsBr phase shows strong bending (the bottom line in Fig. 6). There is no doubt that such a kinetic behaviour is a result of the larger size of the iodine ion entering the CsBr lattice. It may be suggested that such a kinetic behaviour, associated with a smaller rate of establishing the equilibrium when the dissolving atoms or ions are large, will be exhibited by any solid substitutional solution.

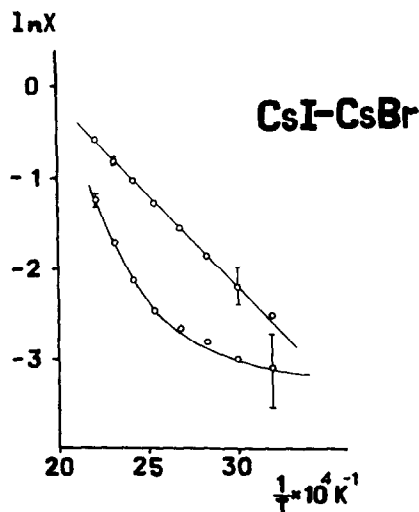


Fig. 6. Vant-Hoff isobars for the solvus curve [5] in the CsI-CsBr system.

† An analogous feature is observed for the equilibrium solvus in the Pb-Sn system (to be published) near the temperature of the structural phase transition of tin ($+140$ – 150°C).

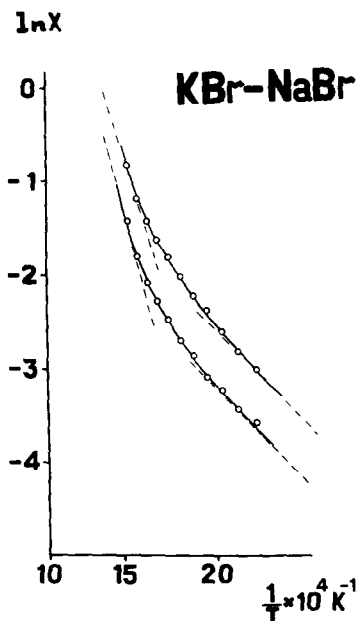


Fig. 7. Vant-Hoff isobars for the solvus branches in the KBr-NaBr system (according to [14]). The dashed line is an approximation of the isobars by straight lines.

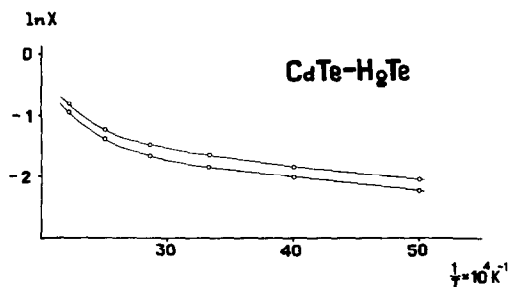


Fig. 8. Vant-Hoff isobars for the solvus branches in the HgTe-CdTe system (according to [13]).

The lower part of Table 1 shows the values of ΔH_S for the solvus curves taken from references whose data were obtained neglecting conditions (1) and (2). Typical examples of isobars (1) constructed on the basis of these data are given in Figs 7 and 8. The solvus for the KBr-NaBr system (Fig. 7) was obtained by the X-ray method using sample quenching [14] and for the HgTe-CdTe system by the e.m.f. method [13]. The non-linearity of the isobars appears to be due in these cases to deviations of the solvus from the equilibrium position. A non-linear isobar

Table 1. The enthalpy of dissolution ΔH_S according to the solvus curves

System	Solution	Temperature interval $\Delta T(K)$	ΔH_S (kcal mol ⁻¹)		$Q_S = \frac{\Delta H_{mix}}{X(1-X)}$ (kcal mol ⁻¹)
			Equilibrium solvus	Solvus from literature data	
KCl-NaCl	NaCl in KCl	498-748	6.7 ± 0.4†	6.7 [10]	4.4 [16]
	KCl in NaCl	498-748	7.4 ± 1.0	6.5	5.5
CsI-CsBr	CsBr in CsI	293-453	4.0 ± 0.4	1.0-2.9 [11]	2.1 [16]
	CsI in CsBr	373-453	3.5 ~ 6.7	1.2-3.9	
Ag-Cu	Cu in Ag	673-1023	9.3 ± 1	9 [1]	
	Ag in Cu	673-1023	17.1 ± 4	17	
Au-Pt	Pt in Au	773-1473	3.3 ± 0.5	4.8 [1]	
	Au in Pt	773-1473	8.5 ± 1	11.1	
Au-Ni	Ni in Au	573-673	4.0 ± 0.5	3.7 [1]	
		773-1073	8.0 ± 1	8	
Pb-Sn	Au in Ni	573-673	3.2	2.2	
		873-1073	19.9 ± 5	8.4-20	
Rh-Cu	Sn in Pb	353-413	3.6 ± 1	3.3 [1]	
	Pb in Sn	413-448	6.3 ± 1	5.9	
HgTe-CdTe	Cu in Rh	413-448	12.4	6.8	
		973-1323	12.1 ± 3		
KBr-NaBr	Rh in Cu	1373-1473	17.7 ± 3		
		873-1173	3.4 ± 0.7		
PbTe-PbS		1173-1473	6.5 ± 1.4		
	PbS in PbTe	673-973		8.6 [12]	
HgTe-CdTe	PbTe in PbS	723-1023		12.5	
	CdTe in HgTe	200-450		0.4-3.1 [13]	
KI-NaI	HgTe in CdTe	200-450		0.45-3.1	
	NaBr in KBr	453-653		5.1-13.3 [14]	3.5 [16]
RbI-RbBr	KBr in NaBr	453-653		6.6-14.8	4.6
	NaI in KI	373-593		3.9-6.6 [14]	2.3 [16]
GaSb-GaAs	KI in NaI	373-593		3.9-6.8	3.2
	RbBr in RbI	217-333		0.7-3.0 [15]	1.6 [16]
	RbI in RbBr	217-333		1.2-2.6	1.8
	GaAs in GaSb	773-973		5.3 [17]	
	GaSb in GaAs	773-973		4.9	

† That is a maximal error of ΔH_S determination received with regard for a maximal error of X composition determination providing the linearity of isobar (1).

Table 2. The validity of eqn (3) for the Au-Pt solid solutions

$t(^{\circ}\text{C})$	X_1 (at.% Pt)	X_2 (at.% Pt)	b
500	15.0	99.4	1.326
600	18.2	98.0	0.963
700	22.8	97.1	1.030
800	26.1	95.0	0.935
900	30.2	93.3	0.954
1000	34.1	91.2	0.960
1100	38.3	89.3	0.992
1200	42.8	85.6	0.977
1300	51.0	82.4	1.069
			$\bar{b} = 0.985$ (600–1300°C) $\sigma_n(b) = 0.041$

can be approximated by straight lines whose slopes allow an estimation of the range of the ΔH_s values (see Fig. 7). These values are listed in Table 1 in the column "solvus from literature data". For solid solutions of the alkali halides the values of ΔH_s are compared with the experimental values of the mutual exchange energy Q_s (a quasi-chemical approximation of the theory of solid solutions [16])

$$Q_s = \frac{\Delta H_{\text{mix}}}{X(1-X)},$$

where ΔH_{mix} is the enthalpy of mixing of the solid solution of composition X according to the calorimetric data. It can be seen from Table 1 that everywhere $\Delta H_s > Q_s$. It may be assumed that one of the reasons for the underestimated values of the calorimetric data is the ageing of the solid solutions.

DISCUSSION

Let us denote

$$h = \frac{\Delta H_{s1}}{\Delta H_{s2}},$$

where ΔH_{s1} is the enthalpy of dissolution of the component with the smaller atomic (ionic) radius in the lattice of the component with the larger radius and ΔH_{s2} is the enthalpy of dissolution of the component with the larger radius in the lattice of the component with the smaller radius. It can be seen from Table 1 that almost always $\Delta H_{s2} > \Delta H_{s1}$. This evidence can be easily explained if one assumes that the main contribution to ΔH_s is the energy of the crystal lattice deformation that takes place upon the isomorphous substitution of the components†. In other words, the value of ΔH_s is

mainly determined by the differences in the dimensions of the atoms of the components, i.e. by the size factor.

It is interesting to note that for the metallic systems Ag-Cu and Au-Pt, as well as for the high-temperature phases of the Au-Ni and Pb-Sn systems, $h = 0.4-0.5$. Such a constancy (if confirmed for other analogous systems) must have a simple explanation within the general principles of the theory of solids. For the salt systems the values of h are higher by approximately a factor of two ($h = 0.7-0.9$, see also Table 3).

As pointed out above, the left-hand branch of the solvus is the solution of the second component (with the smaller atomic radius) in the lattice of the first component (with the larger radius) while the right-hand branch is the solution of the first component in the second. For the left- and the right-hand branches of the solvus we have

$$\begin{aligned} \ln X_1 &= -\frac{\Delta H_{s1}}{RT} + C_1 \\ \ln(1 - X_2) &= -\frac{\Delta H_{s2}}{RT} + C_2, \end{aligned} \quad (2)$$

where indices 1 and 2 correspond to the left- and right-hand branches of the solvus, respectively. To eliminate T let us perform division

$$\frac{\ln X_1 - C_1}{\ln(1 - X_2) - C_2} = \frac{\Delta H_{s1}}{\Delta H_{s2}} = h = \text{const}_1$$

$$\ln X_1 - h \ln(1 - X_2) = C_1 - hC_2$$

$$\ln \frac{X_1}{(1 - X_2)^h} = C_1 - hC_2$$

$$\frac{X_1}{(1 - X_2)^h} = \exp(C_1 - hC_2) = b = \text{const}_2. \quad (3)$$

Equation (3) ("the solvus equation") connects the branches of the solvus for a binary system for which eqn (1) is fulfilled. For the experimental equilibrium solvus curves considered above eqn (3) is well satisfied over a wide range of temperatures. This is illustrated in Table 2 for the case of gold-platinum solid solutions (according to the data of Ref. 7).

It can be seen from Table 2 that eqn (3) is obeyed by this system with an accuracy of about 4% over the temperature range from 600 to 1300°C (the average value of $b = 0.985 \pm \sigma_n(b)$ where $\sigma_n(b) = 0.041$ is the standard deviation). The deviation at $t = 500^{\circ}\text{C}$ is associated with the small value of $1 - X_2$ which considerably increases the relative error in the determination of the composition of the phases. This example illustrates the usefulness of eqn (3) in the analysis of experimental data.

† The Cu-Rh system is an exception. The possible reasons for this will be considered in a separate paper.

Table 3. The parameters of the solvus equation for binary solid solutions

System	Temperature interval (K)	h	$\bar{b}$	$\sigma_s(b)$	X_c (mole fraction of comp. 2)	
					Calc.	Exp.
KCl-NaCl	623-748	0.75	1.532	0.059	0.67	0.63
CsI-CsBr	373-453	0.60	1.186	0.08	0.64	0.64
Ag-Cu	873-1023	0.54	0.608	0.002	—	—
Au-Pt	873-1573	0.42	0.985	0.041	—	—
Au-Ni	873-1073	0.40	1.184	0.091	0.72	0.74
Pb-Sn	413-448	0.51	1.527	0.041	—	—
KBr-NaBr	513-653	0.83	1.287	0.083	0.60	0.65
KI-NaI	393-493	0.83	1.226	0.051	0.59	0.62
RbI-RbBr	217-333	0.86	1.30	0.124	0.59	0.65
PbTe-PbS	723-973	0.67	1.437	0.056	0.67	0.63
HgTe-CdTe	200-450	0.93	1.053	0.021	0.54	0.53
GaSb-GaAs	823-1020	1.08	4.187	0.11	—	—

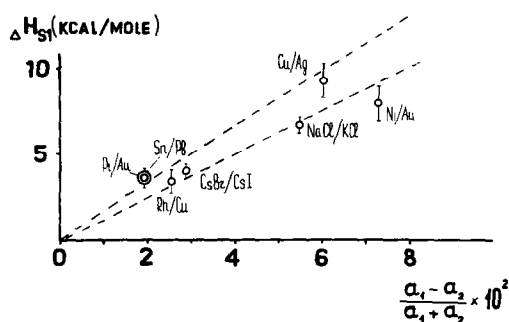


Fig. 9. Dependence of the dissolution enthalpy on the dimensional factor for the equilibrium solvus curves. Designations are given in the text.

For the solvus curve with the critical point of the composition X_c we have at this point that $X_1 = X_2 = X_c$.

Substituting into eqn (3) we obtain

$$\frac{X_c}{(1 - X_c)^h} = b. \quad (4)$$

From this, knowing b and h , we can calculate X_c which is the composition of the solid solution at the critical point. In Table 3 calculated values of X_c are compared with experimental. It can be seen that for the solvus curves obtained without control of the equilibrium state (the lower part of Table 3) the differences between the experimental and calculated values of X_c are greater than for the equilibrium curves (the upper part of the table). The table also shows other parameters of the solvus eqn (3). The values X for the systems in the lower part of Table 3 were taken from the literature sources listed in Table 1. As a result of considerable bending of the isobars of these systems, averaged values of h are given for them.

It is known that in an isomorphous substitution the size factor, i.e. the ratio of the atomic dimensions (or

crystal lattice parameters) of the component of the solid solution, plays an important role. Figure 9 shows the dependence of the enthalpy ΔH_{S1} on the dimensional factor R for the equilibrium solvus curves

$$R = \frac{a_1 - a_2}{a_1 + a_2} \times 10^2,$$

where a_1 and a_2 are the crystal lattice parameters of the components†. It can be seen that this dependence is almost linear. Assuming that

$$\Delta H_{S1} = \lambda_1 R \quad (5)$$

on the basis of Fig. 9 we obtain $\lambda_1 = 1.5 \pm 0.2$ kcal mol⁻¹. The analogous dependence for ΔH_{S2} (the solvent is the component with the smaller atomic radius) shows a considerably greater scatter in the values.

It appears that the values of λ may be somewhat different for systems from different classes (for example, for salt systems, metallic systems and so on). If this assumption is confirmed the estimation of the solubility and construction of the solvus curves will be considerably simplified in many cases.

In conclusion it should be noted that linearity of isobars (1) for the equilibrium solvus curves is rather surprising. This means that the enthalpy of the mutual solubility of the components is constant over a wide range of temperatures (tens and hundreds of degrees). It appears that such a constancy could not be foreseen on the basis of the known facts and theories. In this respect a solid state equilibrium differs significantly from the equilibrium in a solid-liquid system where ΔH_S is known to be strongly temperature-dependent [9].

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† For tin it was assumed that $a = \sqrt[3]{a_T^2 \cdot C_T}$, where a_T and C_T are the parameters of the tetragonal lattice.

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